HERMES QUANT PLATFORM

Institutional Smart Money Concepts & Backtesting Suite

System Specifications & User Documentation

An algorithmic trading companion for Smart Money Concepts (SMC) & ICT setups,
integrated with multi-threaded market scanning, fundamental valuation, and historical strategy validation.

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1. Executive Summary & Overview

The Hermes Quant Platform is an advanced financial analytics and institutional-grade charting application built specifically for retail and professional traders who deploy Smart Money Concepts (SMC) and Inner Circle Trader (ICT) strategies. By bridging the gap between complex algorithmic order-flow analysis and user-friendly visual dashboards, the platform enables traders to identify unmitigated institutional pools, Fair Value Gaps (FVGs), Order Blocks (OBs), and Liquidity Sweeps in real-time.

The platform natively supports the Indian Stock Market (NSE index and component shares), Global Commodities (Gold, Silver, WTI Crude Oil, Natural Gas), and provides developer endpoints to support custom tickers globally via Yahoo Finance. Designed using a sleek dark-themed responsive design, the platform features multi-threaded scanning, real-time Telegram signal integrations, and a lookahead-free Strategy Backtester with custom timeframe resampling.

2. Key Technical Stack

Core UI Framework:	Streamlit (Python-native reactive web application development).
Data Pipeline:	yfinance (Yahoo Finance API client for live and historical data downloads).
Math & Calculations:	Pandas (DataFrame manipulation), NumPy (vectorized operations), and TA-Lib indicators.
Visual Charting:	Plotly (interactive web-based candle charts with custom shape annotations).
Integrations:	Telegram Bot API (automated signaling channels).

3. System Architecture & File Structure

The codebase is structured modularly to separate quantitative analysis from interactive dashboard presentation. Below is a detailed mapping of the platform components:

1. engine.py (Analytics Indicator Engine)

Calculates standard technical indicators (RSI, ATR, VWMA) and runs the custom institutional Smart Money Concept mathematics. Tracks active Fair Value Gaps (FVG) and Order Blocks (OB), and handles dynamic invalidation/mitigation timestamps when price levels violate those boundaries.

2. app.py (User Interface & Controller)

Constructs the Streamlit multi-tab quant interface. Handles layout definitions, sidebar settings, state persistence to protect against tab resets, multi-threaded scanners for Indian Markets and B/S setups, and PDF report generation stylesheets.

3. backtester.py (Strategy Simulation Engine)

Simulates historical performance of the Rule 4 strategy candle-by-candle with zero lookahead bias. Computes win rates, drawdowns, profit factors, and compounded equity curves. Integrates a custom resampling engine to support 3-Hour (3H) and 4-Hour (4H) timeframe testing from 1-Hour base data.

4. telegram_utils.py (Telegram Integration Wrapper)

Dispatches real-time signal alerts and breakout notifications from the setup scanners to designated Telegram channels. Includes a connection verification tool for sidebar setup.

5. nse_list.csv (Ticker Database)

Database mapping NSE stock symbols and index tickers (such as caret-prefixed sector indices) to their full company or index names, utilized by the autocomplete search boxes.

4. Technical Indicator Engine Specifications

The Hermes Indicators Engine (implemented inside engine.py) tracks institutional footprints. Unlike standard moving averages or indicators, SMC focuses on locating price points where large institutions injected high volumes of capital, creating inefficiencies or order collections. The mathematical specs are outlined below:

4.1 Fair Value Gaps (FVG) & Invalidation

A Fair Value Gap is a three-candle price pattern showing a sudden imbalance in market orders.
- Bullish FVG (Demand Zone): Formed when Low(Candle i) > High(Candle i-2). The gap is the range [High(i-2), Low(i)].
- Bearish FVG (Supply Zone): Formed when High(Candle i) < Low(Candle i-2). The gap is the range [High(i), Low(i-2)].
- Mitigation/Invalidation: An FVG is marked active starting at Candle i. It is flagged inactive (closed) and given an 'end' date at the exact timestamp when a future candle's closing price violates or fills the gap (closes below the bottom of a Bull FVG or closes above the top of a Bear FVG).

4.2 Order Blocks (OB) & Dynamic Mitigation

An Order Block is a high-volume candle representing institutional buying or selling blocks.
- Bullish OB (Support Zone): Identified as the last down-close (Bearish) candle before a strong upward expansion that breaks structure. Represented by the high-to-low range of that down candle.
- Bearish OB (Resistance Zone): Identified as the last up-close (Bullish) candle before a strong downward expansion that breaks structure. Represented by the high-to-low range of that up candle.
- Mitigation Invalidation: Order Blocks remain active support/resistance until mitigation occurs. An OB is mitigated and assigned an 'end' timestamp at the exact bar where price closes beyond its boundaries (closes below a Bull OB's bottom or closes above a Bear OB's top). Mitigated OB boxes on the chart cleanly terminate at their mitigation dates to prevent dashboard clutter, leaving only active OB zones extended to the right edge.

5. 'Rule 4' Institutional Setup Strategy

Rule 4 is an institutional confirmation strategy designed to capture high-probability breakouts and retests of key order zones. The strategy utilizes a combination of Order Block breakouts, momentum indicators (RSI), and volume-weighted trend confirmation (VWMA).

5.1 Entry Trigger Rules

To execute a perfect order under the new confluence rules, the following conditions must overlap:

1. Bullish Setup (Long Trades):

- **HTF Confluence: Price must close above the 4H EMA (the bullish gate) to align with the major trend.**
- **Market Structure Shift (MSS): Background/bias must turn Green (candle body closes beyond last swing high).**
- **Liquidity Sweep: Price must sweep below the Green Dashed SSL line at a swing low (low of last 5 bars <= SSL).**
- **Institutional Order Block (OB): Price respects/reacts to a Blue Box Bull OB (OB_Bot < Close <= OB_Top * 1.05).**
- **Fair Value Gap (FVG): An active Green Box Bull FVG must form on the chart.**

2. Bearish Setup (Short Trades):

- **HTF Confluence: Price must close below the 4H EMA (the bearish gate) to filter counter-trend noise.**
- **Market Structure Shift (MSS): Background/bias must turn Red (candle body closes beyond last swing low).**
- **Liquidity Sweep: Price must sweep above the Red Dashed BSL line at a swing high (high of last 5 bars >= BSL).**
- **Institutional Order Block (OB): Price respects/reacts to an Orange Box Bear OB (OB_Bot * 0.95 <= Close < OB_Top).**
- **Fair Value Gap (FVG): An active Red Box Bear FVG must form on the chart.**

5.2 Exit Execution Rules

Once entered, a trade is managed strictly through automated invalidation levels (no human discretion):

1. Long (Buy) Trades:

- **Stop Loss (SL): Set exactly at the bottom boundary of the triggering Bull OB (OB_Bot).**
- **Take Profit (TP): Placed based on a user-defined Risk-to-Reward ratio (R:R), e.g. 1:2.**
  Calculation: TP = Entry + R:R * (Entry - SL).
- **Execution: Exited at SL if candle Low <= SL, or exited at TP if candle High >= TP.**

2. Short (Sell) Trades:

- **Stop Loss (SL): Set exactly at the top boundary of the triggering Bear OB (OB_Top).**
- **Take Profit (TP): Placed based on user-defined Risk-to-Reward ratio.**

Calculation: TP = Entry - R:R * (SL - Entry).

- **Execution: Exited at SL if candle High >= SL, or exited at TP if candle Low <= TP.**

6. Backtesting & Timeframe Resampling

The Strategy Backtester (backtester.py) simulates the performance of Rule 4 over a historical timeline to evaluate win rate, profit factor, equity growth, and drawdowns. A core focus is preventing lookahead bias: indicators and order block mitigation are computed progressively candle-by-candle.

6.1 Multi-Timeframe Resampling Logic (3H, 4H)

Yahoo Finance natively only supports specific intervals (e.g. 1h, 1d) and does not support intermediate timeframes like 3-Hour (3H) or 4-Hour (4H). To support backtesting on these timeframes, the platform:
1. Downloads 1-Hour candles for the user's selected lookback period.
2. Groups and aggregates the candles using Pandas resampling:
- Open = First open price of the block.
- High = Maximum high price of the block.
- Low = Minimum low price of the block.
- Close = Last close price of the block.
- Volume = Sum of volumes in the block.
3. Standardizes the resulting index and feeds it directly into the technical indicators engine.

6.2 Performance Evaluation Metrics

- Win Rate (%): Wins / (Wins + Losses) * 100. (Open trades are excluded from the calculation).
- Profit Factor: Gross Profits / Gross Losses. A value > 1.0 indicates a profitable system.
- Max Drawdown (%): Represents the largest peak-to-trough paper decline in compounded equity.
- Compounded Returns: Simulated starting from 100.0 capital units. Each trade's percentage gain/loss is compounded dynamically to generate a realistic equity curve.

7. User Interface Guide - Module Tabs

The Hermes Quant Dashboard provides a beautiful, responsive multi-tab container. Below is the feature set and layout purpose for each tab:

Tab 1: Main Dashboard

Interactive candlestick chart displaying the selected asset at a locked daily (1d) interval. Renders active Fair Value Gaps, Order Blocks, Liquidity Sweeps, and rolling 50-period VWMA lines. Features theme customization (Dark/Light) and PDF export actions.

Tab 2: Indian Market Scanner

Multi-threaded scanner that processes the top NSE symbols concurrently. Filters assets based on active structure changes and identifies institutional order flow blocks for momentum traders.

Tab 3: Most Probable B/S

Aggregates the top 10 bullish and bearish Rule 4 setups. Leverages ThreadPoolExecutor for high-speed scanning and handles Telegram Bot notifications to alert users instantly when a candidate matches entry rules.

Tab 4: Stock Analysis

Fundamental analyst view containing live stock valuations, health indexes, key metrics, and automatically handles missing fundamental data (e.g. for index ETFs) via generic fallback rules.

Tab 5: Sectorial View

Renders sector matrices and scoreboards. Dynamically uses Pandas map/applymap function depending on the local pandas version to ensure full compatibility without throwing attribute errors.

Tab 6: Commodities

Provides live algorithmic analysis for Gold, Silver, Crude Oil, and Natural Gas using Smart Money Concepts. Features dynamic buy/sell zone mapping and customizable print headers.

Tab 7: Strategy Backtester

Houses the lookahead-free Strategy Backtester. Integrates an autocomplete searchbox (st_searchbox) with automatic symbol validation (e.g. resolving sbin to SBIN.NS) to prevent query failures. Displays high-contrast metrics, a compounded equity curve, detailed trade logs with outcome color-coding, and strategy logic boxes.

8. Deployment & Setup Guide

Follow these steps to deploy and run the Hermes Quant Platform in your local environment:

1. Install Dependencies:
   Ensure Python 3.8+ is installed. Run the command:
   pip install streamlit yfinance pandas numpy plotly TA-Lib streamlit-searchbox fpdf
2. Run the App:
   Execute the command inside the project directory:
   streamlit run app.py
3. Configure Telegram Bot Notifications:
   - Create a Telegram bot by messaging BotFather and copy the Bot Token.
   - Create a channel or group and invite the bot as an administrator.
   - Retrieve the Chat ID (e.g. @my_channel_name).
   - Paste both values into the configurations section in the app sidebar and click 'Test Connection' to verify.